通用方法

Element Software

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通用方法

常用方法是用于检索有关存储集群， API 本身或正在进行的 API 操作的信息的方法。

- GetAPI
- GetAsyncResult
- GetCompleteStats
- GetLimits
- GetOrigin
- GetRawStats
- ListAsyncResults

了解更多信息

- "NetApp SolidFire 资源页面"
- "早期版本的 NetApp SolidFire 和 Element 产品的文档"

GetAPI

您可以使用 `GetAPI` 方法获取可在系统中使用的所有 API 方法和受支持的 API 端点的列表。

Parameters

此方法没有输入参数。

返回值

此方法具有以下返回值：

Name	Description	Type
version	此软件版本支持的所有 API 方法的列表，其中 <version> 是此系统正在运行的当前软件版本。	string array
当前版本	存储集群软件的当前版本。	string
支持的版本	系统支持的所有 API 端点的列表。	string array

请求示例

此方法的请求类似于以下示例：

```
{
  "method": "GetAPI",
  "params": {},
  "id" : 1
}
```

响应示例

此方法返回类似于以下示例的响应：

```
{
"id": 1,
    "result": {
        "12.0": [
            "AbortSnapMirrorRelationship",
            "AddAccount",
            "AddClusterAdmin",
            "AddDrives",
            "AddIdpClusterAdmin",
            "AddInitiatorsToVolumeAccessGroup",
            "AddKeyServerToProviderKmip",
            "AddLdapClusterAdmin",
            "AddNodes",
            "AddVirtualNetwork",
            "AddVolumesToVolumeAccessGroup",
            "BreakSnapMirrorRelationship",
            "BreakSnapMirrorVolume",
            "CancelClone",
            "CancelGroupClone",
            "CheckPingOnVlan",
            "CheckProposedCluster",
            "CheckProposedNodeAdditions",
            "ClearClusterFaults",
            "CloneMultipleVolumes",
            "CloneVolume",
            "CompleteClusterPairing",
            "CompleteVolumePairing",
            "CopyVolume",
            "CreateBackupTarget",
            "CreateClusterInterfacePreference",
            "CreateClusterSupportBundle",
            "CreateGroupSnapshot",
            "CreateIdpConfiguration",
            "CreateInitiators",
```

```
            "CreateKeyProviderKmip",
            "CreateKeyServerKmip",
            "CreatePublicPrivateKeyPair",
            "CreateQoSPolicy",
            "CreateSchedule",
            "CreateSnapMirrorEndpoint",
            "CreateSnapMirrorEndpointUnmanaged",
            "CreateSnapMirrorRelationship",
            "CreateSnapMirrorVolume",
            "CreateSnapshot",
            "CreateStorageContainer",
            "CreateSupportBundle",
            "CreateVolume",
            "CreateVolumeAccessGroup",
            "DeleteAllSupportBundles",
            "DeleteAuthSession",
            "DeleteAuthSessionsByClusterAdmin",
            "DeleteAuthSessionsByUsername",
            "DeleteClusterInterfacePreference",
            "DeleteGroupSnapshot",
            "DeleteIdpConfiguration",
            "DeleteInitiators",
            "DeleteKeyProviderKmip",
            "DeleteKeyServerKmip",
            "DeleteQoSPolicy",
            "DeleteSnapMirrorEndpoints",
            "DeleteSnapMirrorRelationships",
            "DeleteSnapshot",
            "DeleteStorageContainers",
            "DeleteVolume",
            "DeleteVolumeAccessGroup",
            "DeleteVolumes",
            "DisableAutoip",
            "DisableBmcColdReset",
            "DisableClusterSsh",
            "DisableEncryptionAtRest",
            "DisableIdpAuthentication",
            "DisableLdapAuthentication",
            "DisableSnmp",
            "EnableAutoip",
            "EnableBmcColdReset",
            "EnableClusterSsh",
            "EnableEncryptionAtRest",
            "EnableFeature",
            "EnableIdpAuthentication",
            "EnableLdapAuthentication",
```

```
            "EnableSnmp",
            "GetAccountByID",
            "GetAccountByName",
            "GetAccountEfficiency",
            "GetActiveTlsCiphers",
            "GetAsyncResult",
            "GetBackupTarget",
            "GetBinAssignmentProperties",
            "GetClientCertificateSignRequest",
            "GetClusterCapacity",
            "GetClusterConfig",
            "GetClusterFullThreshold",
            "GetClusterHardwareInfo",
            "GetClusterInfo",
            "GetClusterInterfacePreference",
            "GetClusterMasterNodeID",
            "GetClusterSshInfo",
            "GetClusterState",
            "GetClusterStats",
            "GetClusterStructure",
            "GetClusterVersionInfo",
            "GetCompleteStats",
            "GetConfig",
            "GetCurrentClusterAdmin",
            "GetDefaultQoS",
            "GetDriveHardwareInfo",
            "GetDriveStats",
            "GetFeatureStatus",
            "GetFipsReport",
            "GetHardwareConfig",
            "GetHardwareInfo",
            "GetIdpAuthenticationState",
            "GetIpmiConfig",
            "GetIpmiInfo",
            "GetKeyProviderKmip",
            "GetKeyServerKmip",
            "GetLdapConfiguration",
            "GetLimits",
            "GetLldpInfo",
            "GetLoginBanner",
            "GetLoginSessionInfo",
            "GetNetworkConfig",
            "GetNetworkInterface",
            "GetNodeFipsDrivesReport",
            "GetNodeHardwareInfo",
            "GetNodeStats",
```

```
            "GetNtpInfo",
            "GetNvramInfo",
            "GetOntapVersionInfo",
            "GetOrigin",
            "GetPendingOperation",
            "GetProtectionDomainLayout",
            "GetQoSPolicy",
            "GetRawStats",
            "GetRemoteLoggingHosts",
            "GetSSLCertificate",
            "GetSchedule",
            "GetSnapMirrorClusterIdentity",
            "GetSnmpACL",
            "GetSnmpInfo",
            "GetSnmpState",
            "GetSnmpTrapInfo",
            "GetStorageContainerEfficiency",
            "GetSupportedTlsCiphers",
            "GetSystemStatus",
            "GetVirtualVolumeCount",
            "GetVolumeAccessGroupEfficiency",
            "GetVolumeAccessGroupLunAssignments",
            "GetVolumeCount",
            "GetVolumeEfficiency",
            "GetVolumeStats",
            "InitializeSnapMirrorRelationship",
            "ListAccounts",
            "ListActiveAuthSessions",
            "ListActiveNodes",
            "ListActivePairedVolumes",
            "ListActiveVolumes",
            "ListAllNodes",
            "ListAsyncResults",
            "ListAuthSessionsByClusterAdmin",
            "ListAuthSessionsByUsername",
            "ListBackupTargets",
            "ListBulkVolumeJobs",
            "ListClusterAdmins",
            "ListClusterFaults",
            "ListClusterInterfacePreferences",
            "ListClusterPairs",
            "ListDeletedVolumes",
            "ListDriveHardware",
            "ListDriveStats",
            "ListDrives",
            "ListEvents",
```

```
            "ListFibreChannelPortInfo",
            "ListFibreChannelSessions",
            "ListGroupSnapshots",
            "ListISCSISessions",
            "ListIdpConfigurations",
            "ListInitiators",
            "ListKeyProvidersKmip",
            "ListKeyServersKmip",
            "ListNetworkInterfaces",
            "ListNodeFibreChannelPortInfo",
            "ListNodeStats",
            "ListPendingActiveNodes",
            "ListPendingNodes",
            "ListProtectionDomainLevels",
            "ListProtocolEndpoints",
            "ListQoSPolicies",
            "ListSchedules",
            "ListServices",
            "ListSnapMirrorAggregates",
            "ListSnapMirrorEndpoints",
            "ListSnapMirrorLuns",
            "ListSnapMirrorNetworkInterfaces",
            "ListSnapMirrorNodes",
            "ListSnapMirrorPolicies",
            "ListSnapMirrorRelationships",
            "ListSnapMirrorSchedules",
            "ListSnapMirrorVolumes",
            "ListSnapMirrorVservers",
            "ListSnapshots",
            "ListStorageContainers",
            "ListSyncJobs",
            "ListTests",
            "ListUtilities",
            "ListVirtualNetworks",
            "ListVirtualVolumeBindings",
            "ListVirtualVolumeHosts",
            "ListVirtualVolumeTasks",
            "ListVirtualVolumes",
            "ListVolumeAccessGroups",
            "ListVolumeStats",
            "ListVolumeStatsByAccount",
            "ListVolumeStatsByVirtualVolume",
            "ListVolumeStatsByVolume",
            "ListVolumeStatsByVolumeAccessGroup",
            "ListVolumes",
            "ListVolumesForAccount",
```

```
            "ModifyAccount",
            "ModifyBackupTarget",
            "ModifyClusterAdmin",
            "ModifyClusterFullThreshold",
            "ModifyClusterInterfacePreference",
            "ModifyGroupSnapshot",
            "ModifyInitiators",
            "ModifyKeyServerKmip",
            "ModifyQoSPolicy",
            "ModifySchedule",
            "ModifySnapMirrorEndpoint",
            "ModifySnapMirrorEndpointUnmanaged",
            "ModifySnapMirrorRelationship",
            "ModifySnapshot",
            "ModifyStorageContainer",
            "ModifyVirtualNetwork",
            "ModifyVolume",
            "ModifyVolumeAccessGroup",
            "ModifyVolumeAccessGroupLunAssignments",
            "ModifyVolumePair",
            "ModifyVolumes",
            "PurgeDeletedVolume",
            "PurgeDeletedVolumes",
            "QuiesceSnapMirrorRelationship",
            "RemoveAccount",
            "RemoveBackupTarget",
            "RemoveClusterAdmin",
            "RemoveClusterPair",
            "RemoveDrives",
            "RemoveInitiatorsFromVolumeAccessGroup",
            "RemoveKeyServerFromProviderKmip",
            "RemoveNodes",
            "RemoveSSLCertificate",
            "RemoveVirtualNetwork",
            "RemoveVolumePair",
            "RemoveVolumesFromVolumeAccessGroup",
            "ResetDrives",
            "ResetNetworkConfig",
            "ResetNode",
            "ResetSupplementalTlsCiphers",
            "RestartNetworking",
            "RestartServices",
            "RestoreDeletedVolume",
            "ResumeSnapMirrorRelationship",
            "ResyncSnapMirrorRelationship",
            "RollbackToGroupSnapshot",
```

```
            "RollbackToSnapshot",
            "SecureEraseDrives",
            "SetClusterConfig",
            "SetClusterStructure",
            "SetConfig",
            "SetDefaultQoS",
            "SetLoginBanner",
            "SetLoginSessionInfo",
            "SetNetworkConfig",
            "SetNtpInfo",
            "SetProtectionDomainLayout",
            "SetRemoteLoggingHosts",
            "SetSSLCertificate",
            "SetSnmpACL",
            "SetSnmpInfo",
            "SetSnmpTrapInfo",
            "SetSupplementalTlsCiphers",
            "Shutdown",
            "SnmpSendTestTraps",
            "StartBulkVolumeRead",
            "StartBulkVolumeWrite",
            "StartClusterPairing",
            "StartVolumePairing",
            "TestAddressAvailability",
            "TestConnectEnsemble",
            "TestConnectMvip",
            "TestConnectSvip",
            "TestDrives",
            "TestHardwareConfig",
            "TestKeyProviderKmip",
            "TestKeyServerKmip",
            "TestLdapAuthentication",
            "TestLocalConnectivity",
            "TestLocateCluster",
            "TestNetworkConfig",
            "TestPing",
            "TestRemoteConnectivity",
            "UpdateBulkVolumeStatus",
            "UpdateIdpConfiguration",
            "UpdateSnapMirrorRelationship"
        ],
        "currentVersion": "12.0",
        "supportedVersions": [
            "1.0",
            "2.0",
            "3.0",
```

```
            "4.0",
            "5.0",
            "5.1",
            "6.0",
            "7.0",
            "7.1",
            "7.2",
            "7.3",
            "7.4",
            "8.0",
            "8.1",
            "8.2",
            "8.3",
            "8.4",
            "8.5",
            "8.6",
            "8.7",
            "9.0",
            "9.1",
            "9.2",
            "9.3",
            "9.4",
            "9.5",
            "9.6",
            "10.0",
            "10.1",
            "10.2",
            "10.3",
            "10.4",
            "10.5",
            "10.6",
            "10.7",
            "11.0",
            "11.1",
            "11.3",
            "11.5",
            "11.7",
            "11.8",
            "12.0"
        ]
    }
}
```

GetAsyncResult

您可以使用 `GetAsyncResult` 检索异步方法调用的结果。某些方法调用需要一段时间才能运行，并且在系统发送初始响应时可能无法完成。要获取方法调用的状态或结果，请使用 `GetAsyncResult` 轮询方法返回的 asyncHandle 值。

`GetAsyncResult` 以标准方式返回操作的整体状态（正在进行，已完成或错误），但为操作返回的实际数据取决于原始方法调用，并且返回的数据会随每个方法记录在一起。

如果 keepResult 参数缺失或错误，则返回结果后 asyncHandle 将变为非活动状态，并且稍后尝试查询 asyncHandle 时会返回错误。通过将 keepResult 参数设置为 true ，您可以使 asyncHandle 保持活动状态以供将来查询。

Parameters

此方法具有以下输入参数：

Name	Description	Type	默认值	Required
异步处理	从原始异步方法调用返回的值。	整型	无	是的。
keepResult	如果为 true ，则 GetAsyncResult 不会在返回异步结果后将其删除，从而允许将来查询该 asyncHandle 。	boolean	false	否

返回值

此方法具有以下返回值：

Name	Description	Type
status	异步方法调用的状态。可能值： • running ： 此方法仍在运行。 • complete ： 此方法已完成，并且结果或错误可用。	string
结果	如果异步方法成功完成，则这是异步操作的结果。如果异步操作失败，则此成员不存在。	string

Name	Description	Type
error	如果状态为 complete 且异步方法失败，则此成员包含错误详细信息。如果异步操作成功，则此成员不存在。	string
resultType	异步方法调用正在执行或正在执行的操作类型。	string
详细信息	如果状态为 running ，则此成员包含与方法的当前操作相关的信息。如果异步方法未运行，则此成员不存在。	JSON 对象
CREATETIME	调用异步方法的时间，采用 UTC+0 格式。	ISO 8601 日期字符串
最后更新时间	上次更新异步方法状态的时间，采用 UTC+0 格式。	ISO 8601 日期字符串

- 注： * GetAsyncResult 的返回值实质上是标准 JSON 响应的嵌套版本，并具有一个附加的状态字段。

请求示例

此方法的请求类似于以下示例：

```
{
  "method": "GetAsyncResult",
  "params": {
      "asyncHandle" : 389
},
"id" : 1
}
```

响应示例：方法错误

此方法返回类似于以下示例的响应：

```
{
   "error": {
      "code": 500,
      "message": "DBClient operation requested on a non-existent path at
[/asyncresults/1]",
      "name": "xDBNoSuchPath"
   },
   "id": 1
}
```

如果 "response" 是 GetAsyncResult 调用中的 JSON 响应对象，则 "response.error" 将与 GetAsyncResult 方法本身的错误相对应（例如，查询不存在的 asyncHandle ）。

响应示例：异步任务错误

此方法返回类似于以下示例的响应：

```
{
   "id": 1,
   "result": {
     "createTime": "2016-01-01T02:05:53Z",
     "error": {
       "bvID": 1,
       "message": "Bulk volume job failed",
       "name": "xBulkVolumeScriptFailure",
       "volumeID": 34
     },
     "lastUpdateTime": "2016-01-21T02:06:56Z",
     "resultType": "BulkVolume",
     "status": "complete"
   }
}
```

`reonse.result.error` 对应于原始方法调用的错误结果。

响应示例：异步任务成功

此方法返回类似于以下示例的响应：

```
{
   "id": 1,
   "result": {
     "createTime": "2016-01-01T22:29:18Z",
     "lastUpdateTime": "2016-01-01T22:45:51Z",
     "result": {
       "cloneID": 25,
       "message": "Clone complete.",
       "volumeID": 47
     },
     "resultType": "Clone",
     "status": "complete"
   }
}
```

`reonse.result.result` 是在调用成功完成时原始方法调用的返回值。

自版本以来的新增功能

9.6

GetCompleteStats

NetApp 工程人员使用 `GetCompleteStats` API 方法测试新功能。从 `GetCompleteStats` 返回的数据不会进行记录，并且会频繁更改，并且无法保证其准确性。您不应使用 `GetCompleteStats` 收集性能数据或与运行 Element 软件的存储集群进行任何其他管理集成。

使用以下受支持的 API 方法检索统计信息：

- GetVolumeStats
- GetClusterStats
- GetNodeStats
- GetDriveStats

自版本以来的新增功能

9.6

GetLimits

您可以使用 `GetLimits` 方法获取 API 设置的限制值。这些值可能会在 Element 版本之间发生更改，但如果不更新系统，这些值不会发生更改。在为面向用户的工具编写 API 脚本时，了解 API 设置的限制值会很有用。

无论用于传递此方法的 API 端点版本如何， `GetLimits` 方法均返回当前软件版本的限制。

Parameters

此方法没有输入参数。

返回值

此方法将返回一个 JSON 对象，其中包含名称 - 值对，这些名称 - 值对包含 API 限制。

请求示例

此方法的请求类似于以下示例：

```
{
   "method": "GetLimits",
   "id" : 1
}
```

响应示例

此方法返回类似于以下示例的响应：

```
{
    "id": 1,
    "result": {
        "accountCountMax": 5000,
        "accountNameLengthMax": 64,
        "accountNameLengthMin": 1,
        "backupTargetNameLengthMax": 64,
        "backupTargetNameLengthMin": 1,
        "bulkVolumeJobsPerNodeMax": 8,
        "bulkVolumeJobsPerVolumeMax": 2,
        "chapCredentialsCountMax": 15000,
        "cloneJobsPerNodeMax": 8,
        "cloneJobsPerVirtualVolumeMax": 8,
        "cloneJobsPerVolumeMax": 2,
        "clusterAdminAccountMax": 5000,
        "clusterAdminInfoNameLengthMax": 1024,
        "clusterAdminInfoNameLengthMin": 1,
        "clusterPairsCountMax": 4,
        "fibreChannelVolumeAccessMax": 16384,
        "initiatorAliasLengthMax": 224,
        "initiatorCountMax": 10000,
        "initiatorNameLengthMax": 224,
```

```
        "initiatorsPerVolumeAccessGroupCountMax": 128,
        "iscsiSessionsFromFibreChannelNodesMax": 4096,
        "maxAuthSessionsForCluster": 1024,
        "maxAuthSessionsPerUser": 1024,
        "nodesPerClusterCountMax": 100,
        "nodesPerClusterCountMin": 3,
        "qosPolicyCountMax": 500,
        "qosPolicyNameLengthMax": 64,
        "qosPolicyNameLengthMin": 1,
        "scheduleNameLengthMax": 244,
        "secretLengthMax": 16,
        "secretLengthMin": 12,
        "snapMirrorEndpointIPAddressesCountMax": 64,
        "snapMirrorEndpointsCountMax": 4,
        "snapMirrorLabelLengthMax": 31,
        "snapMirrorObjectAttributeValueInfoCountMax": 9900000,
        "snapshotNameLengthMax": 255,
        "snapshotsPerVolumeMax": 32,
        "storageNodesPerClusterCountMin": 2,
        "virtualVolumeCountMax": 8000,
        "virtualVolumesPerAccountCountMax": 10000,
        "volumeAccessGroupCountMax": 1000,
        "volumeAccessGroupLunMax": 16383,
        "volumeAccessGroupNameLengthMax": 64,
        "volumeAccessGroupNameLengthMin": 1,
        "volumeAccessGroupsPerInitiatorCountMax": 1,
        "volumeAccessGroupsPerVolumeCountMax": 64,
        "volumeBurstIOPSMax": 200000,
        "volumeBurstIOPSMin": 100,
        "volumeCountMax": 4000,
        "volumeMaxIOPSMax": 200000,
        "volumeMaxIOPSMin": 100,
        "volumeMinIOPSMax": 15000,
        "volumeMinIOPSMin": 50,
        "volumeNameLengthMax": 64,
        "volumeNameLengthMin": 1,
        "volumeSizeMax": 17592186044416,
        "volumeSizeMin": 1000000000,
        "volumesPerAccountCountMax": 2000,
        "volumesPerGroupSnapshotMax": 32,
        "volumesPerVolumeAccessGroupCountMax": 2000,
        "witnessNodesPerClusterCountMax": 4
    }
}
```

自版本以来的新增功能

9.6

GetOrigin

您可以使用 `GetOrigin` 方法获取构建节点的源证书。

Parameters

> 如果没有源证书，则此方法将返回 "null" 。

此方法没有输入参数。

返回值

此方法将返回供应商原始认证信息。

请求示例

此方法的请求类似于以下示例：

```
{
 "method": "GetOrigin",
 "id" : 1
}
```

响应示例

此方法返回类似于以下示例的响应：

```
{
  "integrator": "SolidFire",
  "<signature>": {
    "pubkey": [public key info],
    "version": 1,
    "data": [signature info]
  },
   "contract-id": "none",
   "location": "Boulder, CO",
   "organization": "Engineering",
   "type": "element-x"
  }
 ]
}
```

自版本以来的新增功能

9.6

GetRawStats

NetApp 工程人员使用 `GetRawStats` API 方法测试新功能。从 `GetRawStats` 返回的数据不会进行记录，并且会频繁更改，并且无法保证其准确性。您不应使用 `GetRawStats` 来收集性能数据或与运行 Element 软件的存储集群进行任何其他管理集成。

使用以下受支持的 API 方法检索统计信息：

- GetVolumeStats
- GetClusterStats
- GetNodeStats
- GetDriveStats

自版本以来的新增功能

9.6

ListAsyncResults

您可以使用 `ListAsyncResults` 列出系统上所有当前正在运行和已完成的异步方法的结果。使用 `ListAsyncResults` 查询异步结果不会使发生原因已完成的 asyncHandles 过期；您可以使用 `GetAsyncResult` 查询 `ListAsyncResults` 返回的任何 asyncHandles 。

参数

此方法具有以下输入参数：

Name	Description	Type	默认值	Required
asyncResultTypes	结果类型的可选列表。您可以使用此列表将结果限制为仅包含这些类型的操作。可能值： • DriveAdd ：涉及系统向集群添加驱动器的操作。 • BulkVolume ：卷之间的复制操作，例如备份或还原。 • clone ：卷克隆操作。 • DriveRemoval ：涉及系统从驱动器复制数据以准备将其从集群中删除的操作。 • RtfiPendingNode ：涉及系统在将兼容软件添加到集群之前在节点上安装该软件的操作。	string array	无	否

返回值

此方法具有以下返回值：

Name	Description	Type
异步处理	序列化异步方法结果数组。	JSON 对象数组

请求示例

此方法的请求类似于以下示例：

```
{
   "method": "ListAsyncResults",
   "params": {
   },
   "id": 1
}
```

响应示例

此方法返回类似于以下示例的响应：

```
{
   "id": 1,
   "result": {
     "asyncHandles": [
       {
         "asyncResultID": 47,
         "completed": true,
         "createTime": "2016-01-01T22:29:19Z",
         "data": {
           "cloneID": 26,
           "message": "Clone complete.",
           "volumeID": 48
         },
         "lastUpdateTime": "2016-01-01T22:45:43Z",
         "resultType": "Clone",
         "success": true
      },
      ...]
   }
}
```

自版本以来的新增功能

9.6

了解更多信息

GetAsyncResult

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